PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Effect of the vancomycin minimum inhibitory concentration on clinical outcomes in patients with methicillin-susceptible Staphylococcus aureus bacteraemia: a systematic review and meta-analysis
AUTHORS	Shi, Changcheng; Ye, Jian; Xu, Renjie; Jin, Weizhong; Xu, Shuang; Teng, Fei; Lin, Nengming

VERSION 1 – REVIEW

REVIEWER	Jennifer McDanel University of Iowa, USA
REVIEW RETURNED	26-Jun-2020

GENERAL COMMENTS	The manuscript describes the association between elevated vancomycin MICs and poor patient outcomes. It is well written and thoroughly conducted. The team extensively performed multiple sub-analyses for the primary outcome of mortally along with exploring a variety of secondary outcomes. I have only a few minor comments and suggestions. 1. In the limitations paragraph on line 271, it mentioned that even though the “sensitivity analysis of the adjusted rate were consistent with the main analysis, other unidentified variables might influence the findings”. Consider listing a few examples of possible variables that could be influencing the findings such as source of infection or specific underlying conditions. Also, perhaps mention that there are differences among included studies in the calculation of severity scores. 2. The article mentions that mainly the included studies calculated 30-day mortality. However, it would be helpful to know more about the variation of the definitions used. Perhaps on Table 1, footnotes could be added to the mortality column describing the definitions used (in-hospital, 30-day, 90-day, etc.).

REVIEWER	James Odei The Ohio State University
REVIEW RETURNED	17-Oct-2020

GENERAL COMMENTS	1. Page 10, Discussion: On line 255 (or line 47-48), remove the word “those”. 2. Page 11, Discussion: On line 267 (or line 13-14), please write “needed” and not “need”.

REVIEWER	Cherry Mahidol-Oxford Tropical Medicine Research Unit (MORU)

REVIEW RETURNED	27-Oct-2020

GENERAL COMMENTS	The findings from this study have potential impact on clinical care of patients with MSSA bacteraemia. The manuscript is well-written and structured. However, there are some concerns in the statistical analysis used. Major comments: All of the 15 studies identified are observational study by design and it is well-known that observational data are prone to different types of biases- including but not limited to confounding bias, selection bias, and measurement bias. It is, therefore, important to critically assess the risk of bias for the 15 studies included in this systematic review. The authors may want to consider ROBINS-I tool for assessing the risk of bias (Sterne JAC, et al. BMJ 2016;355:i4919). The results of the risk of bias assessment for the 15 studies may inform/affect decision on the statistical models to use for the meta-analysis and sensitivity analysis. On Line 145-147, the authors stated that “The fixed effects model was used when significant heterogeneity was not present; otherwise, the random effect model was used.”. The authors should clearly specify which model was used for the results presented. Moreover, authors should be aware that fixed effect and random effects model make different assumptions. Fixed-effect models made an assumption that all studies included in the analysis share a common true effect. In other words, the factors (such as age, clinical severity of the study cohort) influencing the effect size is the same in all the 15 studies. Evidence for this assumption is not supported by Table 1 in the paper. For instance, the comorbidity & disease severity score differed across the 15 studies, suggesting severity of the patient cohort between the studies may not be the same. The authors should specify the evidence that supported the assumption underlying the fixed-effect model whenever used. Please clarify whether the odds ratios for each individual study (Line 143; Figure 1-5) were calculated or directly extracted from the studies, and whether they were adjusted or unadjusted ORs. This is important because unlike randomized controlled trials, observational data are often subjected to different types of biases hence adjusted and unadjusted treatment effects may differ largely. A pooled estimate based on the mixture of the two from different observational studies may introduce more heterogeneity, and is difficult to interpret, and most importantly the validity of such pooled estimate is questionable (https://handbook-5-1.cochrane.org/chapter_13/13_6_2_2_combining_studies.htm). Minor comment: 1. When describing the eligible criteria (Line 122-127), authors could clarify the rationale for the selected study characteristics. For instance, it was not clear why solely paediatric studies were exclude and whether studies with mixed population of adults and children were included or not? If studies of a mixed population of adults and children were included, then whether or not the both adult and paediatric populations were included in the analysis? 2. Please clarify the sentence in Line 136 on “If more than one time point was available, the mortality rate reporting the largest number of deaths was used in the main analysis”. 3. Line 200 in the discussion, please clarify how the findings had supported the statement on evaluated vancomycin MIC correlates

	with a higher mortality rate, “regardless of the antibiotic use”. Were the findings from this study accounted for antibiotic use? 4. Please re-structure the sentence in Line 213-215. 5. Please elaborate further on the potential impact of “selection bias” on the findings of this study (Line 270 in the discussion), and the potential impact of confounding bias (in the studies included) on the findings of this study.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1:
Reviewer Name: Jennifer McDanel
Institution and Country: University of Iowa, USA
1. Comments: The manuscript describes the association between elevated vancomycin MICs and poor patient outcomes. It is well written and thoroughly conducted. The team extensively performed multiple sub-analyses for the primary outcome of mortally along with exploring a variety of secondary outcomes. I have only a few minor comments and suggestions.
1. In the limitations paragraph on line 271, it mentioned that even though the “sensitivity analysis of the adjusted rate were consistent with the main analysis, other unidentified variables might influence the findings”. Consider listing a few examples of possible variables that could be influencing the findings such as source of infection or specific underlying conditions. Also, perhaps mention that there are differences among included studies in the calculation of severity scores.
Response:
Special thanks for the reviewer’s helpful comments. We revised this part of the manuscript according to the reviewer’s comments. On pages 10-11, lines 290-292, “other unidentified variables might influence the findings” were changed to “other unidentified variables (e.g., source of infection, underlying conditions, and the calculation of severity scores) may influence the findings”.

2. Comments: 2. The article mentions that mainly the included studies calculated 30-day mortality. However, it would be helpful to know more about the variation of the definitions used. Perhaps on Table 1, footnotes could be added to the mortality column describing the definitions used (in-hospital, 30-day, 90-day, etc.).
Response:
Special thanks for the reviewer’s helpful comments. According to the reviewer’s suggestion, the definitions of mortality used in the included studies were added in Table 1.

Reviewer 2:
Reviewer Name: James Odei
Institution and Country: The Ohio State University
Comments: Please see attached file for my minor comments.
1. Page 10, Discussion: On line 255 (or line 47-48), remove the word “those”. 2. Page 11, Discussion: On line 267 (or line 13-14), please write “needed” and not “need”.
Response:
Special thanks for the reviewer’s helpful comments. We made corrections according to the comments.
1. On page 10, line 270, “higher than those the values” was changed to “higher than the values”.
2. On page 10, line 282, “need” was changed to “needed”.

Reviewer 3:
Reviewer Name: Cherry
Institution and Country: Mahidol-Oxford Tropical Medicine Research Unit (MORU)
1. Comments:

The findings from this study have potential impact on clinical care of patients with MSSA bacteraemia. The manuscript is well-written and structured. However, there are some concerns in the statistical analysis used.

Major comments:

All of the 15 studies identified are observational study by design and it is well-known that observational data are prone to different types of biases- including but not limited to confounding bias, selection bias, and measurement bias. It is, therefore, important to critically assess the risk of bias for the 15 studies included in this systematic review. The authors may want to consider ROBINS-I tool for assessing the risk of bias (Sterne JAC, et al. BMJ 2016;355:i4919). The results of the risk of bias assessment for the 15 studies may inform/affect decision on the statistical models to use for the meta-analysis and sensitivity analysis.

Response:

Special thanks for the reviewer's helpful comments. According to the recommendation of the Cochrane Collaboration, ROBINS-I tool is the preferred tool used for non-randomized studies of interventions. However, the Cochrane Collaboration noted that the use of the ROBINS-I tool is not mandatory and an alternative option is the Newcastle-Ottawa Scale (NOS) tool (https://methods.cochrane.org/methods-cochrane/robins-i-tool). Therefore, use of the NOS tool for assessing the risk of bias in the original version of the manuscript was reasonable. We believed that the suggestion of the reviewer is very helpful in improving our paper. According to the reviewer's suggestion, the ROBINS-I tool for assessing the risk of bias was used in the revised version of the meta-analysis instead of the NOS tool used in the original version. The sensitivity analysis by restricting studies with low to moderate overall risk of bias was also performed. The results did not alter the main findings.

On page 5, lines 133-134, "The Newcastle-Ottawa Scale (NOS) was used to assess the quality of observational studies 11" was changed to "The Risk of Bias in Nonrandomized Studies for Interventions (ROBINS-I) tool was used to assess the quality of observational studies 11".

On page 6, line 154, "high-quality (NOS score ≥ 7)" was changed to "high-quality (low to moderate risk of bias in 'overall assessment' of ROBIN-I tool)".

On pages 6-7, lines 174-176, "The NOS scores ranged from 5 to 8 points (Table S2)" was changed to "Risk of bias assessment showed that six studies 13,15,18,22-24 had an overall critical risk of bias, and nine studies 8,9,14,16,17,19-21,25 had an overall low to moderate risk of bias (Table S6).".

On page 14, lines 379-380, "11. Stang A. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. Eur J Epidemiol 2010;25: 603-5." was changed to "11. Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ. 2016;355:i4919.".

Table 3, the results of sensitivity analysis of studies with low to moderate overall risk of bias are presented.

Table S6 was added to the supplemental data.

2. Comments: On Line 145-147, the authors stated that "The fixed effects model was used when significant heterogeneity was not present; otherwise, the random effect model was used.". The authors should clearly specify which model was used for the results presented. Moreover, authors should be aware that fixed effect and random effects model make different assumptions. Fixed-effect models made an assumption that all studies included in the analysis share a common true effect. In other words, the factors (such as age, clinical severity of the study cohort) influencing the effect size is the same in all the 15 studies. Evidence for this assumption is not supported by Table 1 in the paper. For instance, the comorbidity & disease severity score differed across the 15 studies, suggesting severity of the patient cohort between the studies may not be the same. The authors should specify the evidence that supported the assumption underlying the fixed-effect model whenever used.

Response:

Special thanks for the reviewer's helpful comments. According to the reviewer's suggestion, the models used for the results are presented in the revised version of the meta-analysis. For the

outcomes using fixed effects model, we also calculated the random effects model for additional analyses to provide conservative pooling estimates. The results did not alter the main findings.
On page 6, lines 148-150, the following text was added: “For the outcomes using fixed effects model, we also calculated the random effects model for additional analyses to provide conservative pooling estimates.”.
On page 7, lines 179-180, the following text was added: “the main analysis using a fixed effects model showed”.
On page 7, lines 182-183, the following text was added: “The additional analysis using the random effects model showed similar results (OR, 1.52; 95% CI, 1.08 to 2.12; I2=40.3%) (Table 2).”
On page 7, line 193, the following text was added: “in the pooled analysis using the random effects model”.
On page 7, line 195, the following text was added: “using the fixed effects model”.
On page 7, lines 196-197, the following text was added: “The additional analysis using the random effects model supported the main conclusions (Table 2).”.
On page 7, line 199, the following text was added: “in the pooled analysis using the random effects model”.
On page 8, lines 206-207, the following text was added: “The results of the additional analyses using the random effects model supported the main conclusions (Table 2).”
Table 2, the results of the additional analyses using the random effects model were added and the overall main analyses were added. A column of “Effects model” was also added.

3. Comments: Please clarify whether the odds ratios for each individual study (Line 143; Figure 1-5) were calculated or directly extracted from the studies, and whether they were adjusted or unadjusted ORs. This is important because unlike randomized controlled trials, observational data are often subjected to different types of biases hence adjusted and unadjusted treatment effects may differ largely. A pooled estimate based on the mixture of the two from different observational studies may introduce more heterogeneity, and is difficult to interpret, and most importantly the validity of such pooled estimate is questionable (https://handbook-5-1.cochrane.org/chapter_13/13_6_2_2_combining_studies.htm).
Response:
Special thanks for the reviewer’s helpful comments. We revised this part of the manuscript according to the reviewer’s comments. To clarify whether the odds ratios for each individual study were calculated or directly extracted from the studies and whether they were adjusted or unadjusted ORs, relevant information is presented in Tables S2-S5.
On page 6, lines 173-174, the following text was added: “Raw data for outcomes extracted from the included studies are presented in Tables S2-S5.”
I agree with the reviewer that a pooled estimate based on the mixture of the two from different observational studies may introduce more heterogeneity, and it is difficult to interpret. Most importantly, the validity of this pooled estimate is questionable. According to the reviewer’s comments, separate meta-analyses were performed for unadjusted and adjusted prognostic effect estimates to further confirm the robustness of our findings. Similar results were observed between pooled unadjusted and adjusted estimates, which indicates the robustness of our findings.
On pages 5-6, lines 144-146, the following text was added: “If adjusted ORs were provided in the included studies, then they were used for data combination. Otherwise, unadjusted ORs were used for data combination.”
On page 6, line 155, “adjusted mortality rate” was changed to “unadjusted mortality rate, adjusted mortality rate”.
On page 7, lines 183-184, the following text was added: “14 studies providing unadjusted mortality rates (OR, 1.42; 95% CI, 1.13 to 1.79; I2=36.0%),”.
Table 3, the results of sensitivity analysis of unadjusted mortality are presented.
Tables S2-S5 were added in the supplemental data.

4. Comments:
Minor comment: 1. When describing the eligible criteria (Line 122-127), authors could clarify the rationale for the selected study characteristics. For instance, it was not clear why solely paediatric studies were exclude and whether studies with mixed population of adults and children were included or not? If studies of a mixed population of adults and children were included, then whether or not the both adult and paediatric populations were included in the analysis?
Response:
Special thanks for the reviewer's helpful comments. According to the reviewer's suggestion, we carefully checked the patient characteristics, especially the age. Among the 15 studies in the meta-analysis, adult patients solely were included in nine studies (Aguado 2011, Baxi 2016, Bouiller 2018, Fernandez-Hidalgo 2018, Han 2012, Lopez-Cortes 2015, San-Juan 2016, Song 2017, and Sullivan 2017). Mixed populations of adults and children were included in two studies (Falcon 2019 and Holmes 2011). However, only 9 children (< 8%) were included in the study by Falcon et al,. and 4.1% were < 18 years of age in the study by Holmes et al. It is unclear whether any children were included in the remaining four studies based on the inclusion criteria (Abelenda 2018, Caston 2014, Cervera 2014 and Videma 2014). The exact proportion of children in these studies was also unavailable. However, the patient characteristics of these studies showed the mean or median age was more than 60 years.
Overall, the mean or median age in the included studies ranged from 60 to 68 years, as we described in RESULTS section, which indicates that the population was dominated by older aged adults. Considering appreciable differences between adults and children in physiology and disease management, solely paediatric studies were excluded in the current meta-analysis. We believed that it is justifiable from an internal validity and a generalizability point of view.

5. Comments: 2. Please clarify the sentence in Line 136 on "If more than one time point was available, the mortality rate reporting the largest number of deaths was used in the main analysis".
Response:
Thank you for these helpful comments. We made corrections according to the comments.
Page 5, lines 136-138, "If more than one time point was available, the mortality rate reporting the largest number of deaths was used in the main analysis" was changed to "When data from more than one time point were available, the mortality at the latest point reporting the largest number of deaths was used in the main analysis (e.g., 30-day mortality had precedence over 14-day mortality)."

6. Comments: 3. Line 200 in the discussion, please clarify how the findings had supported the statement on evaluated vancomycin MIC correlates with a higher mortality rate, "regardless of the antibiotic use". Were the findings from this study accounted for antibiotic use?
Response:
Special thanks for the reviewer's helpful comments. As non-native English speakers, we have a language barrier to the phrase "regardless of". Regarding the sentence "regardless of the antibiotic use", what we really want to express is "do not consider the impact of the antibiotic use". As we mentioned in the INTRODUCTION section, the choice of antibiotic is closely related to the survival of patients with MSSA bacteraemia. We tried our best to evaluated the potential impact of the antibiotic use to the conclusions. However, as we stated in the limitations, the dose and duration of antibiotic therapy and the types of β-lactams chosen were not reported in most included studies. Due to insufficient details of empirical and definitive treatments, a subgroup analysis based on the antibiotic treatment could not be performed.
To avoid ambiguity, the following text was removed: ", regardless of the antibiotic used" (page 8, line 215).

7. Comments: 4. Please re-structure the sentence in Line 213-215.
Response:
Special thanks for the reviewer's helpful comments.

On page 8, lines 228-229, “Similar results were reported in the study by Cervera et al., which included patients who exclusively received cloxacillin 17, included in this meta-analysis.” was changed to “Similar results were reported in the included study by Cervera et al., which included patients who exclusively received cloxacillin 17.”

8. Comments: 5. Please elaborate further on the potential impact of “selection bias” on the findings of this study (Line 270 in the discussion), and the potential impact of confounding bias (in the studies included) on the findings of this study.
Response:
Special thanks for the reviewer’s helpful comments. We revised this part of the manuscript according to the reviewer’s comments.
On page 10, lines 285-289, “First, only data from observational studies were available on this topic; thus, selection bias likely affects the estimate of the effect size.” were changed to “First, all studies included in the meta-analysis were observational studies containing many confounding factors. Therefore, selection bias and comparability bias likely affected the estimate of the effect size. For example, patients with high vancomycin MICs had numerically higher mean or median CCI score in several studies 8,17,22. Bouiller et al. observed more dialysis and higher proportion ultimately fatal disease in the high vancomycin MICs group 15.”

We have done our best to improve the manuscript and made changes in the manuscript, but these changes did not influence the content or framework of the paper. The changes are marked in red in the revised paper, and they are listed here:
1. On page 4, line 106, “8,9.” were changed to “8,9.”
2. On page 7, line 188, “(Table S3)” were changed to “(Table S7)”.
3. On page 7, lines 190-191, “Secondary outcome data were available in nine studies 8,13,15-18,21,22,24 and the definitions were shown in Table S4” were changed to “Secondary outcome data were available in nine studies 8,13,15-18,21,22,24, and the definitions are shown in Table S8”.
4. On page 12, lines 320-324, “NM” were changed to “Nengming Lin”, “all authors” were changed to “Changcheng Shi, Jian Ye, Renjie Xu, Weizhong Jin, Shuang Xu, Fei Teng, and Nengming Lin”, “CS” were changed to “Changcheng Shi”, and “All the authors” were changed to “Changcheng Shi, Jian Ye, Renjie Xu, Weizhong Jin, Shuang Xu, Fei Teng, and Nengming Lin”.
5. On page 15, line 413, “1057-1066” were changed to “1057-66”.
6. On page 17, line 469, “Included” were changed to “included”.
7. On page 17, lines 482-485, the following text was added: “
Table S2 Raw data of mortality extracted from the included studies Table S3 Raw data of septic thrombophlebitis extracted from the included studies Table S4 Raw data of persistent bacteraemia extracted from the included studies Table S5 Raw data of complicated bacteraemia extracted from the included studies”.
8. On page 17, line 486, “Table S2” were changed to “Table S6”.
9. On page 17, line 487, “Table S3” were changed to “Table S7”.
10. On page 17, line 486, “Table S4” were changed to “Table S8”.
11. Table 1, “Falcon [18]” were changed to “Falcon 2019 [18]”; “228” were changed to “230”; mortality rate with different definitions were added in several included studies (Baxi 2016, Cervera 2014, Lopez-Cortes 2015, Sullivan 2017, and Viedma 2014).
12. Table S8, “Falcon [18]” were changed to “Falcon 2019 [18]”; “associated to” were changed to “associated with”; “more than” were changed to “>”; “a duration of > 3 days” were changed to “a duration > 3 days”.

We sincerely appreciate the work of the editors/reviewers and hope that the corrections we have made will be met with approval.

VERSION 2 – REVIEW

REVIEWER	Jennifer McDanel University of Iowa, USA
REVIEW RETURNED	08-Dec-2020

GENERAL COMMENTS	No additional comments to the authors.

REVIEWER	James Odei The Ohio State University, USA.
REVIEW RETURNED	28-Nov-2020

GENERAL COMMENTS	The paper is well written. The concerns or comments made by reviewers were appropriately addressed. So, I do not have any further comments at this time.

REVIEWER	Cherry Lim Mahidol-Oxford Tropical Medicine Research Unit (MORU)
REVIEW RETURNED	09-Dec-2020

GENERAL COMMENTS	I have three minor comments on the revised manuscript: 1. In the Discussion section on limitations, the first point that the authors mentioned is related to “confounding bias”. This is good and certainly is an important discussion point. However, it is unclear how this is related to the “selection bias and comparability bias likely affected” the estimated effect size. Please elaborate. I would suggest the authors to elaborate the influence of “confounding bias” on the findings by highlighting that only 6 out of the 15 studies identified and included in this study reported an adjusted odds ratio and also highlight the findings from the subgroup analyses of the current study. “Selection bias” could be discussed as another limitation with better example. The example on “patients with high vancomycin MICs had numerically higher mean or median CCI score in several studies” seems to suggest CCI score is a potential confounder (which likely that is the case) that should be adjusted but was not adjusted in many studies identified. How this example is related to bias introduced by the selection of individuals in the studies identified was not immediately obvious in the discussion. Just to point out that confounding bias and selection bias are two different types of biases. Confounding bias is a result of having confounder(s) and failed to adjust for them properly. Selection bias is introduced by the way participants are selected into the study and analysis. Moreover, please elaborate the term “comparability bias”. Is this a new terminology? If yes, please provide references. 2. Adding symbol (i.e. *) directly in the Figure 2 with the appropriate footnote (i.e. text to explain the symbol) to highlight which “OR (95% CI)” are crude OR and which ones are adjusted OR would be very useful for readers. 3. The format of the new Table 2 is slightly odd and took some time for me to digest. A suggestion would be to remove the column with the heading “Effect model”, and re-word the heading on “Main

	analyses" (i.e. as "Estimates from fixed effect models"). Also re-word the heading on "Additional analyses using the random effect effects model" (i.e. to "Estimates from random effect models"). Also, maybe "Not performed/done" is more appropriate than "Not available" for the column on OR, and "Not applicable" is more appropriate for the column for P value in Table 2.

VERSION 2 – AUTHOR RESPONSE

Reviewer 2:
Reviewer Name: James Odei
Institution and Country: The Ohio State University
Comments: The paper is well written. The concerns or comments made by reviewers were appropriately addressed. So, I do not have any further comments at this time.
Response: Thank you.

Reviewer 1:
Reviewer Name: Jennifer McDanel
Institution and Country: University of Iowa, USA
Comments: No additional comments to the authors.
Response: Thank you.

Reviewer 3:
Reviewer Name: Cherry
Institution and Country: Mahidol-Oxford Tropical Medicine Research Unit (MORU)
1. Comments: I have three minor comments on the revised manuscript: 1. In the Discussion section on limitations, the first point that the authors mentioned is related to "confounding bias". This is good and certainly is an important discussion point. However, it is unclear how this is related to the "selection bias and comparability bias likely affected" the estimated effect size. Please elaborate.

I would suggest the authors to elaborate the influence of "confounding bias" on the findings by highlighting that only 6 out of the 15 studies identified and included in this study reported an adjusted odds ratio and also highlight the findings from the subgroup analyses of the current study.

"Selection bias" could be discussed as another limitation with better example. The example on "patients with high vancomycin MICs had numerically higher mean or median CCI score in several studies" seems to suggest CCI score is a potential confounder (which likely that is the case) that should be adjusted but was not adjusted in many studies identified. How this example is related to bias introduced by the selection of individuals in the studies identified was not immediately obvious in the discussion. Just to point out that confounding bias and selection bias are two different types of biases. Confounding bias is a result of having confounder(s) and failed to adjust for them properly. Selection bias is introduced by the way participants are selected into the study and analysis.

Moreover, please elaborate the term "comparability bias". Is this a new terminology? If yes, please provide references.

Response:
Special thanks for the reviewer's helpful comments. We revised this part of the manuscript according to the reviewer's comments.
On pages 10-11, lines 285-299, "First, all studies included in the meta-analysis were observational studies containing many confounding factors. Therefore, selection bias and comparability bias likely

affected the estimate of the effect size. For example, patients with high vancomycin MICs had numerically higher mean or median CCI score in several studies 8,17,22. Bouiller et al. obseved more dialysis and higher proportion ultimately fatal disease in the high vancomycin MICs group 15. Although the results of the sensitivity analysis of the adjusted mortality rate were consistent with the main analysis, other unidentified variables (e.g., source of infection, underlying conditions, and the calculation of severity scores) may influence the findings." were changed to "First, all studies included in the meta-analysis were observational studies containing many confounding factors. However, only six out of the included studies were designed to adjust for confounding factors and reported adjusted mortality rates. For studies without adjustment for potential confounders, the baseline characteristics of the patients (e.g., underlying conditions and the calculation of severity scores) between the high and low vancomycin MIC groups may influence the outcomes. For example, patients with high vancomycin MICs had numerically higher mean or median CCI scores in several studies 17,22. Bouiller et al. observed more dialysis and a higher proportion of ultimately fatal disease in the high vancomycin MIC group 15. More underlying diseases and higher CCI scores might be associated with adverse outcomes. Therefore, the adverse effect of the high vancomycin MIC on clinical outcomes might be overestimated. Second, almost half of the studies were retrospective in design, which may introduce inherent selection bias into this report. To address the potential confounding bias and selection bias, numerous sensitivity analyses were performed. Sensitivity analyses limited to prospective studies, high-quality studies based on the ROBIN-I tool, and studies providing adjusted mortality data did not substantially alter the results of the main analysis, further supporting the conclusions in our study."
The term "comparability bias" comes from the Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomized studies (Stang A. Eur J Epidemiol 2010;25: 603-5). The NOS contains eight items, categorized into three dimensions: selection, comparability, and outcome (cohort studies) or exposure (case-control studies). Comparability bias refers to adjustment for potential covariates, similar to the concept of confounding bias. The term "comparability bias" has been used in several meta-analyses (PMID: 20625008, 26123860 and 30983822). However, the term "confounding bias" seems to be more commonly used than the term "comparability bias". Therefore, we replaced the term "comparability bias" with the term "confounding bias" in the revised version.
Once again, thank you very much for your helpful comments and suggestions.

2. Comments: 2. Adding symbol (i.e. *) directly in the Figure 2 with the appropriate footnote (i.e. text to explain the symbol) to highlight which "OR (95% CI)" are crude OR and which ones are adjusted OR would be very useful for readers.
Response:
Special thanks for the reviewer's helpful comments. To highlight which "OR (95% CI)" are crude OR and which ones are adjusted OR, we added the column with the heading "Adjusted OR" in Figures 2-5. We declare that the data in the figures have not been modified.

3. Comments: 3. The format of the new Table 2 is slightly odd and took some time for me to digest. A suggestion would be to remove the column with the heading "Effect model", and re-word the heading on "Main analyses" (i.e. as "Estimates from fixed effect models"). Also re-word the heading on "Additional analyses using the random effect effects model" (i.e. to "Estimates from random effect models"). Also, maybe "Not performed/done" is more appropriate than "Not available" for the column on OR, and "Not applicable" is more appropriate for the column for P value in Table 2.

Response:
Special thanks for the reviewer's helpful comments. We revised Table 3 according to the reviewer's comments. We declare that the data in the Table 3 has not been modified.

We have done our best to improve the manuscript and made changes in the manuscript, but these changes did not influence the content or framework of the paper. The changes are marked in red in

the revised paper, and they are listed here:
1. On page 11, line 299, “Second” was changed to “Third”.
2. On page 11, line 306, “Third” was changed to “Fourth”.
3. On page 13, line 354, “methicillinresistant” was changed to “methicillin-resistant”.

We sincerely appreciate the work of the editors/reviewers and hope that the corrections we have made will be met with approval.

VERSION 3 – REVIEW

REVIEWER	Cherry Lim Mahidol-Oxford Tropical Medicine Research Unit (MORU), Bangkok, Thailand
REVIEW RETURNED	20-Dec-2020

GENERAL COMMENTS	All comments are appropriately addressed by the authors. I have no additional comments.